Saikannan Sathish

(607) 621-8860 | ssathish2@binghamton.edu | LinkedIn | Github | Portfolio

EDUCATION

Binghamton University, State University of NY, Thomas J. Watson College of Engineering and Applied Science
Master of Science in Computer Science (Artificial Intelligence) *Expected Aug 2026*
Cumulative GPA: 3.7/4.00

Sri Ramachandra University, Sri Ramachandra Faculty of Engineering and Technology
Bachelor of Technology in Computer Science Engineering (Artificial Intelligence and Machine Learning) *July 2024*
Cumulative GPA: 3.69/4.00

PROFESSIONAL EXPERIENCE

Courser AI, Founding Engineer | Cursor for Curriculum (AI Startup) | USA *January 2026 - Present*
- Architected and deployed a full-stack AI-powered course planning platform using React, Flask, PostgreSQL, Railways
- Integrated OpenRouter and Google Gemini APIs for AI-driven syllabus generation with real-time editing and validation
- Onboarded Happy Academy (10+ teachers, 50+ students), iterated on teacher feedback to ship bulk CSV import, resource management and enrollment workflows

Uplifty AI, Machine Learning Engineer Lead | Texas, USA *August 2025 - Present*
- Built a personalized social media recommendation system leveraging user interaction data and behavioral signals to rank posts for new and existing users
- Developed scalable backend APIs using Python, FastAPI, Supabase, PostgreSQL and deployed on Google Cloud Run
- Implemented a multi-factor ranking algorithm using engagement, recency decay, safety filtering and growth signals
- Led end-to-end feature development for the Content Feed module, coordinating cross-team integration

Sellularity, Founding Engineer | sellularity.org | USA *January 2025 - May 2025*
- Engineered a platform for a de-identified healthcare data exchange, carried product from idea to implementation
- Utilized healthcare interoperability standards and Epic EHR smart Protocol based to construct secure ETL pathways for patients healthcare data. Backend: Flask, Python. Infrastructure: AWS, KMS, RDS, ECS. Frontend: Angular
- Achieved interview round for Ycombinator and presented technical demo to investors.

Verzeo, AIML Engineer Intern | India *December 2022 - February 2023*
- Developed a real-time hand gesture recognition model using TensorFlow and OpenCV
- Implemented object detection API, improving gesture recognition accuracy by 92%
- Utilized OpenCV, TensorFlow, LabelImg and the Object Detection API in the development process
- Collaborated cross-functionally to integrate the model into production workflows, aligning with stakeholder needs

Qurinom Solutions, Mobile App Developer Intern | India *January 2022 - March 2022*
- Architectured and developed a full-stack app using Flutter and Firebase with authentication and state management
- Designed Firestore schema and integrated backend workflows with MySQL and PostgreSQL exposure

PROJECT EXPERIENCE

Gitwise - AI Codebase Intelligence | Independent Project | Link *Jan 2026 - April 2026*
- Architected a 6-node LangGraph state machine (clone, embed, architecture, security, quality, readme) that analyzes any GitHub repository and streams real-time reports via SSE, achieving 30-90s end-to-end analysis
- Implemented hybrid RAG pipeline using Supabase pgvector cosine similarity and filepath keyword retrieval, enabling natural-language question and answer over codebases with exact file and line citations via Groq LLaMA-3.3-70B
- Built regex-based security scanner (hardcoded secrets, SQL injection, eval misuse - CRITICAL to LOW severity) and code quality grader (cyclomatic complexity, test coverage, dead-code detection, A-F grade)
- Engineered production reliability features: exponential backoff retry, 2-hour session TTL with background cleanup, cold-start warmup detection and client-side ZIP report export

Hybrid Knowledge Assistant: Full-Stack RAG System | Independent Project | Github *Oct 2025 - Present*
- Built a modular RAG pipeline using FastAPI, Supabase (pgvector) supporting document ingestion, chunking, embedding and semantic retrieval for PDF/TXT/URL sources
- Developed REST endpoints (/ingest, /ask) for scalable document indexing and citation-aware LLM-based Q&A
- Designed a lightweight frontend with file upload, chat interface and configurable retrieval parameters for experimentation and evaluation.

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C, C++, Dart, SQL
AI/ML & LLM: RAG, LangGraph, LangChain, Groq API, HuggingFace, LLMs, TensorFlow, PyTorch, Scikit-learn, Recommendation Systems, NLP, Pandas, NumPy
Backend & APIs: FastAPI, Flask, REST API, Async Processing, Server-Sent Event (SSE), OpenRouter API, Gemini API
Frontend & Mobile: React.js, Next.js, Flutter, Tailwind CSS, HTML5, CSS3
Databases: PostgreSQL, MySQL, MongoDB, Firebase, Supabase, pgvector
Core CS: Data Structures & Algorithms, Object-Oriented Programming
Cloud & Infrastructure: Git, CI/CD, Vercel, Render, Google Cloud Run, AWS (KMS, RDS, ECS), Docker, Azure, AWS